Mission Project Timeline

JoanSat

Organization:	DPI Research
Satellite Class:	CubeSat
Spacecraft Mass:	1.0 kg
Target Launch:	May 2029
Document Date:	2026-05-02

1. Executive Summary

This document presents a comprehensive project timeline for the **JoanSat** mission, a CubeSat with a target launch date of May 2029. The critical path analysis estimates **31 months** from project initiation to launch readiness (with parallel activities), compared to **36 months** available. This provides a schedule margin of **5 months**. The overall probability of on-time completion is assessed at **92%** (favorable outlook). 10 compatible launch vehicles were identified.

Metric	Value
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Schedule Margin	+5 months
On-Time Probability	92%
Compatible Launch Vehicles	10

Table 1 — Timeline Key Metrics

2. Project Timeline (Gantt Chart)

The following chart shows all project phases from concept through launch. Solid bars represent nominal durations; dashed extensions show pessimistic estimates. The red dashed line marks the target launch date.

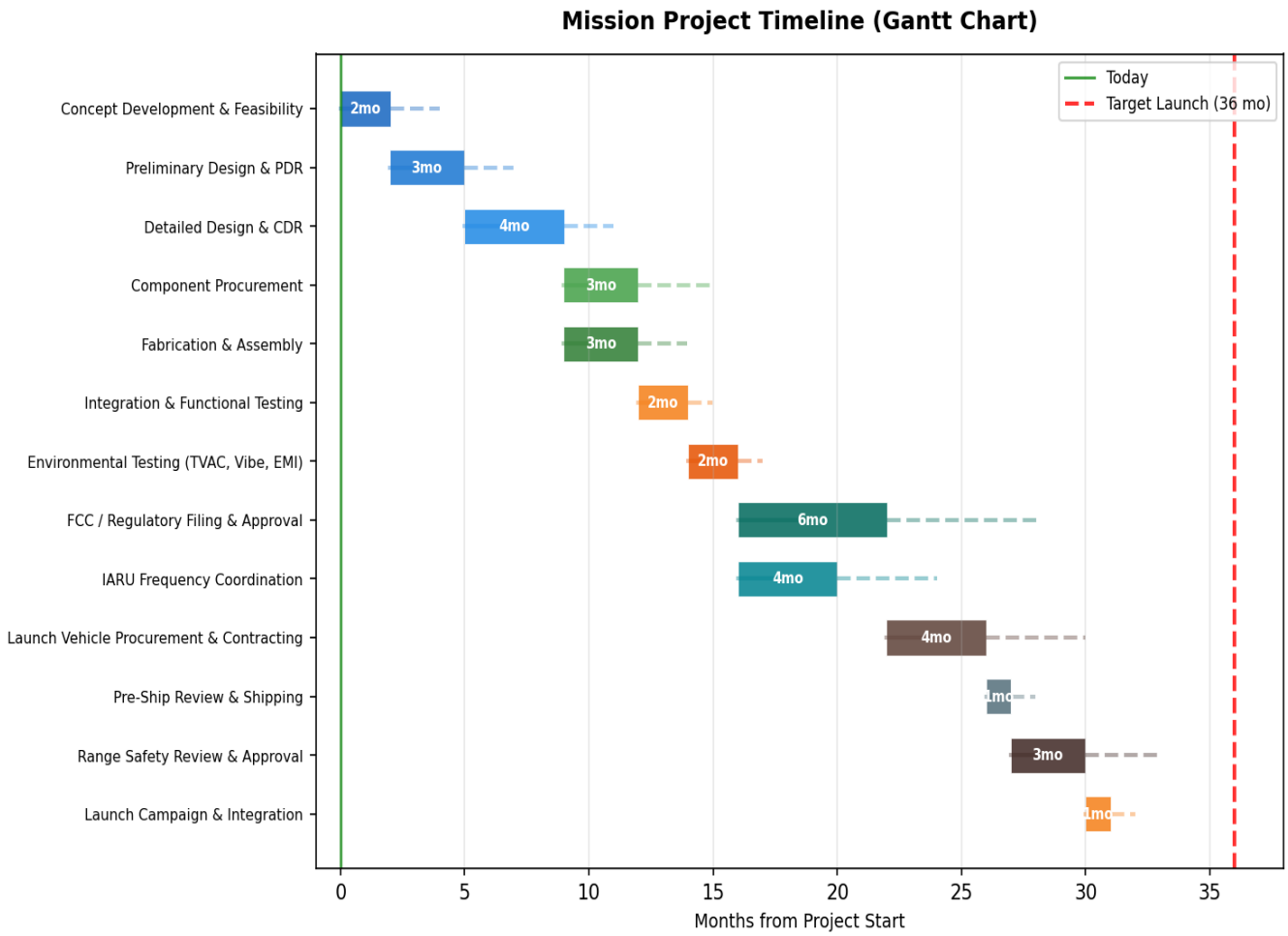

Figure 1 — Mission Project Gantt Chart

3. Phase Details

Each phase includes optimistic, nominal, and pessimistic duration estimates based on industry data for the satellite class and mission parameters.

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Table 2 — Phase Schedule Detail

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4. Launch Vehicle Analysis

Based on the spacecraft mass (1 kg) and target orbit (Sun-Synchronous (SSO)), the following launch vehicles are compatible:

Vehicle	Operator	Capacity	Cost ($M)	Lead Time	Status
D-Orbit ION (Orbital Transfer)	D-Orbit	150 kg	$0.5M	6 mo	Operational
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Table 3 — Compatible Launch Vehicles

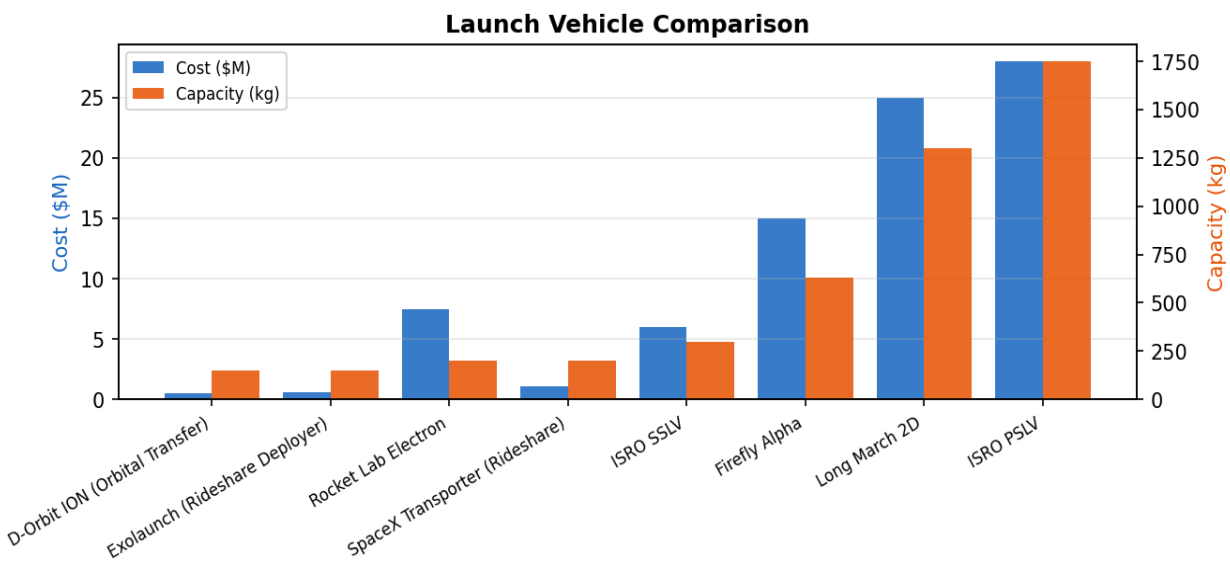

Figure 2 — Launch Vehicle Cost & Capacity Comparison

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D-Orbit ION (Orbital Transfer) by D-Orbit is the top-ranked match based on capacity fit, operational status, and cost efficiency. It offers 150 kg to the target orbit (99% mass margin). Estimated launch cost is $0.5M. Typical procurement lead time is 6 months. Notes: Last-mile delivery vehicle. Deploys smallsats to precise orbits from rideshare.

5. Probability of On-Time Completion

The assessed probability of launching on or before May 2029 is **92%** (favorable).

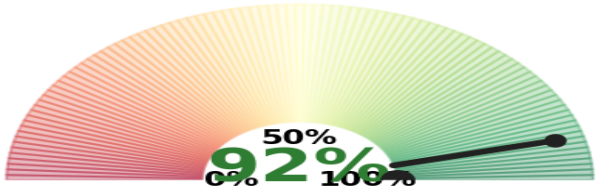

Figure 3 — On-Time Completion Probability

5.1 Risk Factor Analysis

The probability assessment considers the following factors. Positive values increase confidence; negative values decrease it.

Factor	Impact	Assessment
Schedule Margin	+4	5 months of margin (16% buffer) — adequate but tight
Satellite Complexity	+5	CubeSat — well-understood class with COTS components, short timelines
Launch Vehicle Availability	+5	13 operational vehicles available — excellent launch flexibility
Regulatory Path	-2	Standard Part 25.114 — full FCC review process, typically 12+ months

Table 4 — Probability Risk Factors

6. Recommendations

• **Regulatory Filing:** Initiate FCC filing and frequency coordination in parallel with preliminary design. Regulatory approvals are often on the critical path and have unpredictable timelines.

7. Assumptions & Caveats

• Phase durations are based on industry averages for the selected satellite class and may vary significantly based on team experience, funding profile, and supply chain conditions.

• Parallel phase execution assumes adequate staffing and facilities to support concurrent activities.

• FCC processing times are estimates; actual timelines depend on application completeness, FCC backlog, and any requests for additional information.

• Launch vehicle costs and availability are based on publicly available data as of 2025 and are subject to change. Actual pricing requires a formal quote from the launch provider.

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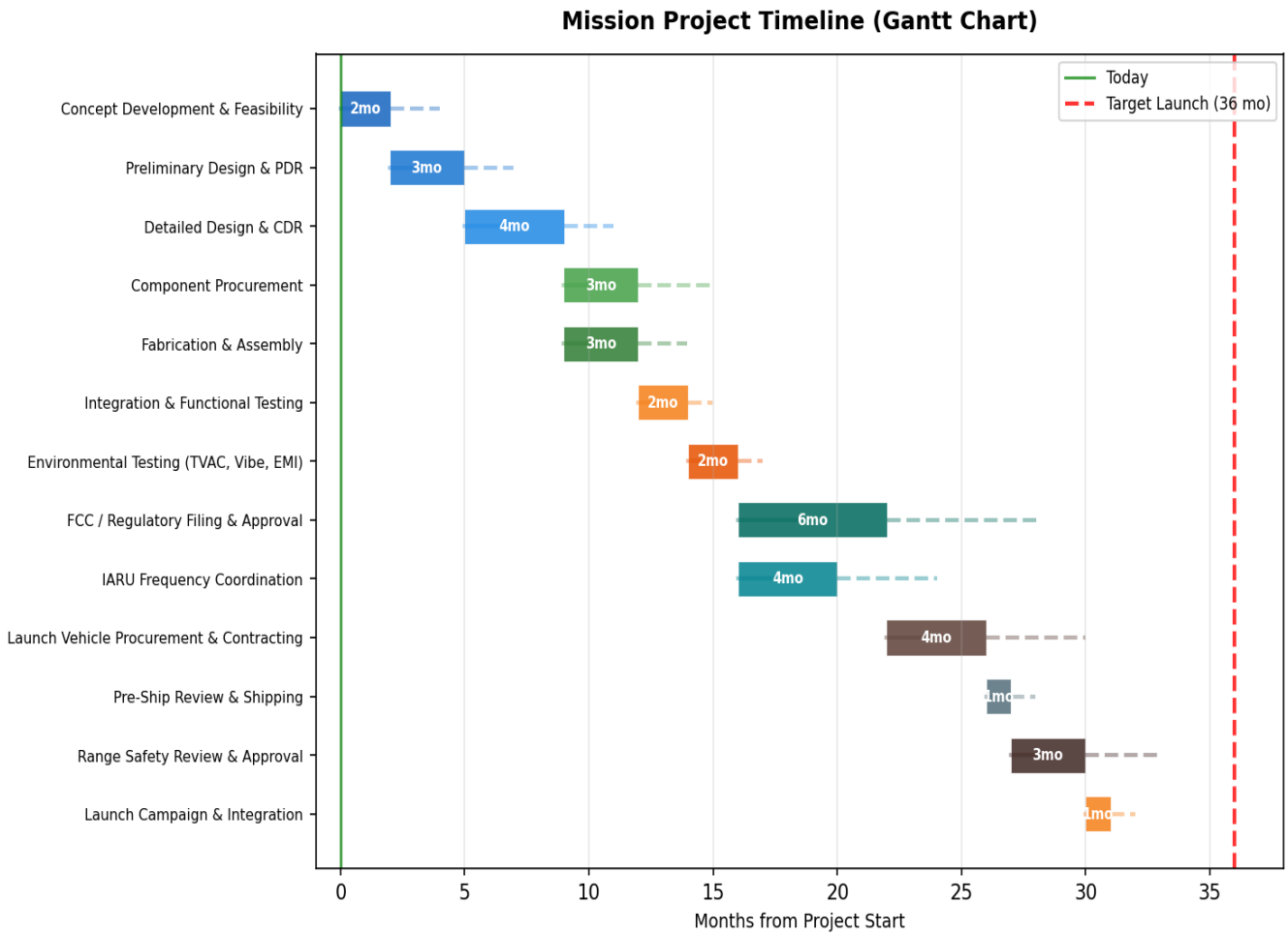

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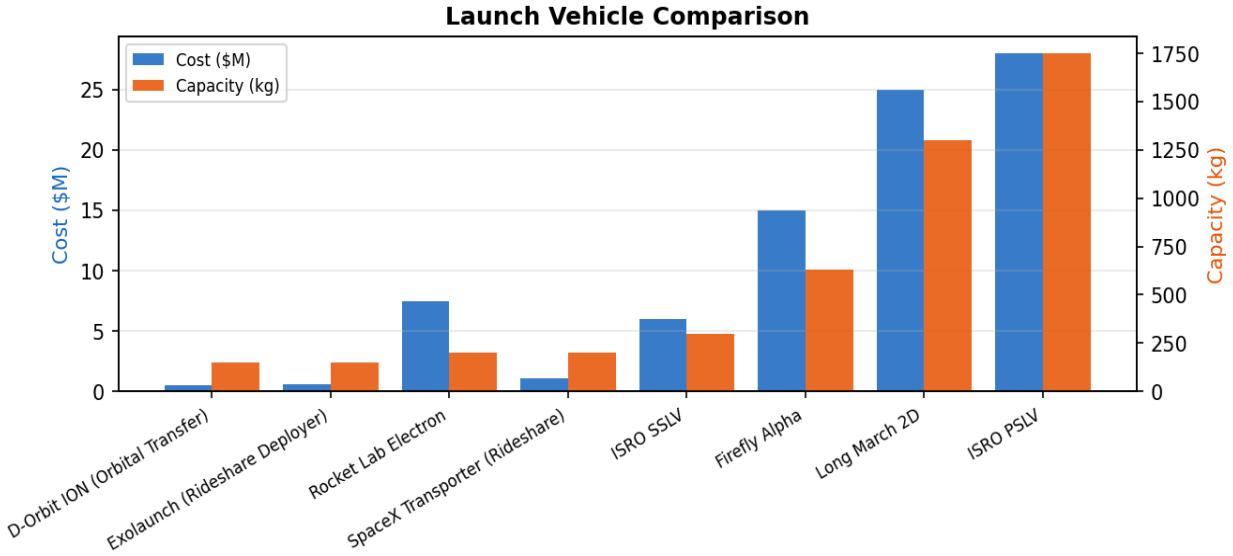

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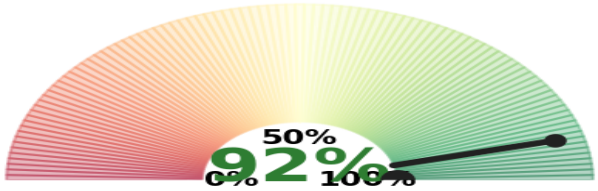

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